

MODULE-6

INNER PRODUCT SPACES

4.1 INTRODUCTION

Inner product space is a vector space with an inner product on it. It associates each pair of vectors in the space with a scalar quantity known as the inner product of vectors. It helps defining the orthogonality between vectors. They generalize Euclidean spaces to the vector spaces of any dimension.

4.2 INNER PRODUCT SPACES

Let V be a real vector space. An inner product on V denoted by $\langle \cdot, \cdot \rangle$ is a function from $V \times V \rightarrow \mathbb{R}$ that assigns a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ to each ordered pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in V in such a way that for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V and all scalars k , the following axioms are satisfied.

- (a) $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ (Symmetry)
- (b) $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$ (Additivity)
- (c) $\langle k\mathbf{u}, \mathbf{v} \rangle = k \langle \mathbf{u}, \mathbf{v} \rangle$ (Homogeneity)
- (d) $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ (non-negativity)

if and only if $\mathbf{u} = \mathbf{0}$

If the given product satisfies all the above 4 axioms then V is called a real inner product space with respect to the given product.

Note: If $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ are vectors in $\mathbb{R}^n$ then Euclidean inner product (dot product)

4.2.1 *Properties of Inner Products*

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in an inner product space V and k is any scalar then

$$(i) \quad \langle \mathbf{0}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{0} \rangle = 0$$

$$(ii) \quad \langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$$

$$(iii) \quad \langle \mathbf{u}, k\mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle$$

$$(iv) \quad \langle \mathbf{u} - \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle - \langle \mathbf{v}, \mathbf{w} \rangle$$

$$(v) \quad \langle \mathbf{u}, \mathbf{v} - \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{w} \rangle$$

Example 1: Let $\mathbf{u} = (3, -1)$, $\mathbf{v} = (2, -2)$, $\mathbf{w} = (-1, 6)$ and $k = -2$. Verify the following using Euclidean inner product.

$$(i) \quad \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$$

$$(ii) \quad \langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$$

$$(iii) \quad \langle k\mathbf{u}, \mathbf{v} \rangle = k \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, k\mathbf{v} \rangle$$

$$(iv) \quad \langle \mathbf{0}, \mathbf{u} \rangle = \langle \mathbf{u}, \mathbf{0} \rangle = 0$$

Solution: (i) $\mathbf{u} + \mathbf{v} = (3, -1) + (2, -2) = (5, -3)$

$$\text{L.H.S.} = \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle$$

$$= (\mathbf{u} + \mathbf{v}) \cdot \mathbf{w}$$

$$= (5, -3) \cdot (-1, 6)$$

$$= -5 - 18$$

$$= -23$$

$$\text{R.H.S.} = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$$

$$= \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$$

$$= (3, -1) \cdot (-1, 6) + (2, -2) \cdot (-1, 6)$$

$$= (-3 - 6) + (-2 - 12)$$

$$= -23$$

$$\text{L.H.S.} = \text{R.H.S.}$$

$$\text{(iii)} \quad k\mathbf{u} = -2(3, -1) = (-6, 2), \quad k\mathbf{v} = -2(2, -2) = (-4, 4)$$

$$\langle k\mathbf{u}, \mathbf{v} \rangle = (k\mathbf{u}) \cdot \mathbf{v}$$

$$= (-6, 2) \cdot (2, -2)$$

$$= (-12 - 4)$$

$$= -16$$

$$k\langle \mathbf{u}, \mathbf{v} \rangle = k(\mathbf{u} \cdot \mathbf{v})$$

$$= -2[(3, -1) \cdot (2, -2)]$$

$$= -2(6 + 2)$$

$$= -16$$

$$\langle \mathbf{u}, k\mathbf{v} \rangle = \mathbf{u} \cdot (k\mathbf{v})$$

$$= (3, -1) \cdot (-4, 4)$$

$$= -12 - 4$$

$$= -16$$

$$\langle k\mathbf{u}, \mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, k\mathbf{v} \rangle$$

Example 2: Determine which of the following are inner products on R^3 if, $\mathbf{u} = (u_1, u_2, u_3)$, $\mathbf{v} = (v_1, v_2, v_3)$

(i) $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + u_2v_2 + 4u_3v_3$

(ii) $\langle \mathbf{u}, \mathbf{v} \rangle = u_1^2v_1^2 + u_2^2v_2^2 + u_3^2v_3^2$

(iii) $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 - u_2v_2 + u_3v_3$

Solution: (i) (a) $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + u_2v_2 + 4u_3v_3$

$$= 2v_1u_1 + v_2u_2 + 4v_3u_3$$
$$= \langle \mathbf{v}, \mathbf{u} \rangle$$

Symmetry axiom is satisfied.

(b) Let $\mathbf{w} = (w_1, w_2, w_3)$ be also in R^3 .

$$\mathbf{u} + \mathbf{v} = (u_1, u_2, u_3) + (v_1, v_2, v_3)$$
$$= (u_1 + v_1, u_2 + v_2, u_3 + v_3)$$

$$\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = 2(u_1 + v_1)w_1 + (u_2 + v_2)w_2 + 4(u_3 + v_3)w_3$$
$$= (2u_1w_1 + u_2w_2 + 4u_3w_3) + (2v_1w_1 + v_2w_2 + 4v_3w_3)$$
$$= \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$$

(c) Let k be any scalar.

$$\begin{aligned}k\mathbf{u} &= k(u_1, u_2, u_3) = (ku_1, ku_2, ku_3) \\ \langle k\mathbf{u}, \mathbf{v} \rangle &= 2(ku_1)v_1 + (ku_2)v_2 + 4(ku_3)v_3 \\ &= k(2u_1v_1 + u_2v_2 + 4u_3v_3) \\ &= k\langle \mathbf{u}, \mathbf{v} \rangle\end{aligned}$$

Homogeneity axiom is satisfied.

$$\begin{aligned}\text{(d)} \quad \langle \mathbf{u}, \mathbf{u} \rangle &= 2u_1u_1 + u_2u_2 + 4u_3u_3 \\ &= 2u_1^2 + u_2^2 + 4u_3^2 \geq 0\end{aligned}$$

$$\text{Also, } \langle \mathbf{u}, \mathbf{u} \rangle = 2u_1^2 + u_2^2 + 4u_3^2 = 0$$

if and only if $u_1 = 0, u_2 = 0, u_3 = 0$ i.e. $\mathbf{u} = \mathbf{0}$

Non-negativity axiom is satisfied.

Hence, the given product is an inner product in R^3 .

$$\begin{aligned}
 \text{(ii) (a) } \langle \mathbf{u}, \mathbf{v} \rangle &= u_1^2 v_1^2 + u_2^2 v_2^2 + u_3^2 v_3^2 \\
 &= v_1^2 u_1^2 + v_2^2 u_2^2 + v_3^2 u_3^2 \\
 &= \langle \mathbf{v}, \mathbf{u} \rangle
 \end{aligned}$$

Symmetry axiom is satisfied.

(b) Let $\mathbf{w} = (w_1, w_2, w_3)$ be also in R^3 .

$$\begin{aligned}
 \mathbf{u} + \mathbf{v} &= (u_1, u_2, u_3) + (v_1, v_2, v_3) \\
 &= (u_1 + v_1, u_2 + v_2, u_3 + v_3)
 \end{aligned}$$

$$\begin{aligned}
 \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle &= (u_1 + v_1)^2 w_1^2 + (u_2 + v_2)^2 w_2^2 + (u_3 + v_3)^2 w_3^2 \\
 &= (u_1^2 w_1^2 + u_2^2 w_2^2 + u_3^2 w_3^2) + (v_1^2 w_1^2 + v_2^2 w_2^2 + v_3^2 w_3^2) + 2(u_1 v_1 w_1^2 + u_2 v_2 w_2^2 + u_3 v_3 w_3^2) \\
 &= \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle + 2(u_1 v_1 w_1^2 + u_2 v_2 w_2^2 + u_3 v_3 w_3^2) \\
 &\neq \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle
 \end{aligned}$$

Additivity axiom is failed.

Hence, the given product is not an inner product in R^3 .

Example 4: If $\mathbf{u} = (u_1, u_2)$, $\mathbf{v} = (v_1, v_2)$ are vectors in R^2 then prove that R^2 is an inner product space with respect to the inner product defined as

$$\langle \mathbf{u}, \mathbf{v} \rangle = 4u_1v_1 + u_2v_1 + 4u_1v_2 + 4u_2v_2.$$

4.2.3 Norm or Length in Inner Product Spaces

The norm or length of a vector $\mathbf{u}$ in an inner product space V is denoted by $\|\mathbf{u}\|$ and is defined by

$$\|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{\frac{1}{2}}$$

Unit Vector: Let $\mathbf{u}$ be a vector in an inner product space V . If $\|\mathbf{u}\| = 1$ then $\mathbf{u}$ is called a unit vector in V .

Properties of Length

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in an inner product space V and k is any scalar then

1. $\|\mathbf{u}\| \geq 0$
2. $\|\mathbf{u}\| = 0$ if and only if $\mathbf{u} = 0$
3. $\|k\mathbf{u}\| = |k| \|\mathbf{u}\|$
4. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$, Triangle inequality.

4.2.4 *Distance in Inner Product Spaces*

The distance between two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space V is denoted by $d(\mathbf{u}, \mathbf{v})$ and is defined by

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

Properties of Distance

If $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are vectors in an inner product space V then

1. $d(\mathbf{u}, \mathbf{v}) \geq 0$
2. $d(\mathbf{u}, \mathbf{v}) = 0$ if and only if $\mathbf{u} = \mathbf{v}$
3. $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$
4. $d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$, Triangle inequality

Example 1: Find $\|\mathbf{u}\|$ if $\mathbf{u} = (3, 4)$ and weighted Euclidean inner product is $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 - u_1 v_2 - u_2 v_1 + 3u_2 v_2$ where $\mathbf{u} = (u_1, u_2)$, $\mathbf{v} = (v_1, v_2)$.

Solution:

$$\begin{aligned}\|\mathbf{u}\| &= \langle \mathbf{u}, \mathbf{u} \rangle^{\frac{1}{2}} \\ &= (u_1 u_1 - u_1 u_2 - u_2 u_1 + 3u_2 u_2)^{\frac{1}{2}} \\ &= (3^2 - 3 \cdot 4 - 4 \cdot 3 + 3(4)^2)^{\frac{1}{2}} \\ &= \sqrt{33}.\end{aligned}$$

Example 7: Find $\langle \mathbf{f}, \mathbf{g} \rangle$ if $\mathbf{f} = f(x) = 1 - x + x^2 + 5x^3$, $\mathbf{g} = g(x) = x - 3x^2$ and the inner product $\langle \mathbf{f}, \mathbf{g} \rangle = \int_{-1}^1 f(x)g(x) dx$

Solution: $\langle \mathbf{f}, \mathbf{g} \rangle = \int_{-1}^1 (1 - x + x^2 + 5x^3)(x - 3x^2) dx$

$$= \int_{-1}^1 (x - 3x^2 - x^2 + 3x^3 + x^3 - 3x^4 + 5x^4 - 15x^5) dx$$

$$= \int_{-1}^1 (x - 4x^2 + 4x^3 + 2x^4 - 15x^5) dx$$

$$= 2 \int_0^1 (-4x^2 + 2x^4) dx \left[\begin{array}{l} \because \int_{-a}^a f(x) dx = 0, \text{ if } f(x) \text{ is odd} \\ = 2 \int_0^a f(x) dx, \text{ if } f(x) \text{ is even} \end{array} \right]$$

$$= 2 \left| -4 \frac{x^3}{3} + 2 \frac{x^5}{5} \right|_0^1$$

$$= 2 \left(-\frac{4}{3} + \frac{2}{5} \right)$$

$$= -\frac{28}{15}.$$

Example 8: Find $d(\mathbf{f}, \mathbf{g})$ if $\mathbf{f} = f(x) = \cos 2\pi x$ and $\mathbf{g} = g(x) = \sin 2\pi x$ and the inner product $\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^1 f(x)g(x) dx$

Solution: $d(\mathbf{f}, \mathbf{g}) = \|\mathbf{f} - \mathbf{g}\|$

$$= \langle \mathbf{f} - \mathbf{g}, \mathbf{f} - \mathbf{g} \rangle^{\frac{1}{2}}$$

$$\langle \mathbf{f} - \mathbf{g}, \mathbf{f} - \mathbf{g} \rangle = \int_0^1 [f(x) - g(x)][f(x) - g(x)] dx$$

$$= \int_0^1 (\cos 2\pi x - \sin 2\pi x)^2 dx$$

$$= \int_0^1 (\cos^2 2\pi x + \sin^2 2\pi x - 2 \cos 2\pi x \sin 2\pi x) dx$$

$$= \int_0^1 (1 - \sin 4\pi x) dx$$

$$= \left[x + \frac{\cos 4\pi x}{4\pi} \right]_0^1$$

$$= 1 + \frac{\cos 4\pi - \cos 0}{4\pi}$$

$$= 1$$

$$d(\mathbf{f}, \mathbf{g}) = \sqrt{1} = 1.$$

4.2.5 Angle between Vectors

If $\mathbf{u}$ and $\mathbf{v}$ are non-zero vectors in an inner product space V and if θ is the angle between them then

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

4.2.6 Orthogonality

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space V are called orthogonal if

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0 \quad \text{i.e.,} \quad \theta = \frac{\pi}{2}$$

4.2.7 Pythagorean Theorem

If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal vectors in an inner product space V then

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

Proof: Since $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle = 0$

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} + \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \quad \left[\because \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle = 0 \right] \end{aligned}$$

4.2.8 Cauchy–Schwarz Inequality

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in an inner product space V then

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

Example 1: Find the cosine of the angle between $\mathbf{u}$ and $\mathbf{v}$ if R^2 , R^3 and R^4 have the Euclidean inner product.

- (i) $\mathbf{u} = (1, -3), \mathbf{v} = (2, 4)$
- (ii) $\mathbf{u} = (-1, 5, 2), \mathbf{v} = (2, 4, -9)$
- (iii) $\mathbf{u} = (1, 0, 1, 0), \mathbf{v} = (-3, -3, -3, -3)$

Solution: $\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}$, where θ is angle between $\mathbf{u}$ & $\mathbf{v}$.

$$= \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

$$\begin{aligned} \text{(i) } \cos \theta &= \frac{(1, -3) \cdot (2, 4)}{\sqrt{1+9} \sqrt{4+16}} \\ &= \frac{2-12}{\sqrt{10} \sqrt{20}} \\ &= \frac{-10}{\sqrt{10} \sqrt{20}} \\ &= -\frac{1}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} \text{(ii) } \cos \theta &= \frac{(-1, 5, 2) \cdot (2, 4, -9)}{\sqrt{1+25+4} \sqrt{4+16+81}} \\ &= \frac{-2+20-18}{\sqrt{30} \sqrt{101}} \\ &= 0 \end{aligned}$$

Example 4: Find cosine of the angle between $\mathbf{p}_1 = x - x^2$, and $\mathbf{p}_2 = 7 + 3x + 3x^2$ if the inner product $\langle \mathbf{p}_1, \mathbf{p}_2 \rangle = a_0b_0 + a_1b_1 + a_2b_2$, where

$$\mathbf{p}_1 = a_0 + a_1x + a_2x^2 \text{ and } \mathbf{p}_2 = b_0 + b_1x + b_2x^2.$$

Solution: Let θ be the angle between $\mathbf{p}_1$ and $\mathbf{p}_2$.

$$\begin{aligned}\cos \theta &= \frac{\langle \mathbf{p}_1, \mathbf{p}_2 \rangle}{\|\mathbf{p}_1\| \|\mathbf{p}_2\|} \\&= \frac{\langle x - x^2, 7 + 3x + 3x^2 \rangle}{\sqrt{(1)^2 + (-1)^2} \sqrt{(7)^2 + (3)^2 + (3)^2}} \\&= \frac{(0)(7) + (1)(3) + (-1)(3)}{\sqrt{2} \sqrt{67}} \\&= 0\end{aligned}$$

Example 2: Determine whether the given vectors are orthogonal with respect to the Euclidean inner product.

- (i) $\mathbf{u} = (-1, 3, 2), \mathbf{v} = (4, 2, -1)$
- (ii) $\mathbf{u} = (-4, 6, -10, 1), \mathbf{v} = (2, 1, -2, 9)$

Solution: (i) $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v}$

$$\begin{aligned} &= (-1, 3, 2) \cdot (4, 2, -1) \\ &= (-1)(4) + (3)(2) + (2)(-1) \\ &= -4 + 6 - 2 \\ &= 0 \end{aligned}$$

Hence, $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

(ii) $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v}$

$$\begin{aligned} &= (-4, 6, -10, 1) \cdot (2, 1, -2, 9) \\ &= (-4)(2) + (6)(1) + (-10)(-2) + (1)(9) \\ &= -8 + 6 + 20 + 9 \\ &= 27 \neq 0 \end{aligned}$$

Hence, $\mathbf{u}$ and $\mathbf{v}$ are not orthogonal.

Example 3: Determine whether there exists scalars k and l such that the vectors $\mathbf{u} = (2, k, 6)$, $\mathbf{v} = (l, 5, 3)$ and $\mathbf{w} = (1, 2, 3)$ are mutually orthogonal with respect to the Euclidean inner product.

Solution: Let $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ be mutually orthogonal.

$$\begin{aligned}\langle \mathbf{u}, \mathbf{v} \rangle &= \mathbf{u} \cdot \mathbf{v} = 0 \\ (2, k, 6) \cdot (l, 5, 3) &= 0 \\ 2l + 5k + 18 &= 0 \\ 2l + 5k &= -18 \quad \dots(1)\end{aligned}$$

$$\begin{aligned}\langle \mathbf{v}, \mathbf{w} \rangle &= \mathbf{v} \cdot \mathbf{w} = 0 \\ (l, 5, 3) \cdot (1, 2, 3) &= 0 \\ l + 10 + 9 &= 0 \\ l &= -19 \quad \dots(2)\end{aligned}$$

$$\begin{aligned}\langle \mathbf{w}, \mathbf{u} \rangle &= \mathbf{w} \cdot \mathbf{u} = 0 \\ (1, 2, 3) \cdot (2, k, 6) &= 0 \\ 2 + 2k + 18 &= 0 \\ k &= -10 \quad \dots(3)\end{aligned}$$

Substituting l and k in equation (1),

$$\begin{aligned}2(-19) + 5(-10) &= -18 \\ -88 &= -18\end{aligned}$$

This shows that l and k does not satisfy equation (1).

Hence, there do not exist k and l such that $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are orthogonal.

Example 5: Show that $\mathbf{p}_1 = 1 - x + 2x^2$ and $\mathbf{p}_2 = 2x + x^2$ are orthogonal with respect to the inner product $\langle \mathbf{p}_1, \mathbf{p}_2 \rangle = a_0b_0 + a_1b_1 + a_2b_2$, where

$$\mathbf{p}_1 = a_0 + a_1x + a_2x^2 \text{ and } \mathbf{p}_2 = b_0 + b_1x + b_2x^2.$$

Solution:

$$\begin{aligned}\langle \mathbf{p}_1, \mathbf{p}_2 \rangle &= \langle 1 - x + 2x^2, 2x + x^2 \rangle \\ &= (1)(0) + (-1)(2) + (2)(1) \\ &= 0\end{aligned}$$

Hence, $\mathbf{p}_1$ and $\mathbf{p}_2$ are orthogonal.

Example 6: Show that the matrices $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ are orthogonal with respect to the inner product $\langle A, B \rangle = a_1b_1 + a_2b_2 + a_3b_3 + a_4b_4$ where

$$A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \text{ and } B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}.$$

Solution:

$$\begin{aligned}\langle A, B \rangle &= (2)(1) + (1)(1) + (-1)(0) + (3)(-1) \\ &= 0\end{aligned}$$

Hence, A and B are orthogonal.

4.2.9 Orthogonal and Orthonormal Set

A set $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ of vectors in an inner product space V is called an orthogonal set if each pair of distinct vectors in S are orthogonal, i.e. $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for $i \neq j$.

An orthogonal set of unit vectors (norm is 1) is called orthonormal, i.e. $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for $i \neq j$ and $\langle \mathbf{u}_i, \mathbf{u}_i \rangle = 1$ for $i = 1, 2, \dots, p$.

The process of dividing a non-zero vector $\mathbf{u}$ by its norm is called normalizing $\mathbf{u}$.

$$\text{normalized } \mathbf{u} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$$

Example 1: Show that the vectors $\mathbf{u}_1 = \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}\right)$, $\mathbf{u}_2 = \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right)$ and $\mathbf{u}_3 = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$ are orthonormal with respect to the Euclidean inner product on R^3 .

Solution:

$$\begin{aligned}\langle \mathbf{u}_1, \mathbf{u}_2 \rangle &= \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}\right) \cdot \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right) \\ &= \left(\frac{2}{3}\right)\left(\frac{2}{3}\right) + \left(-\frac{2}{3}\right)\left(\frac{1}{3}\right) + \frac{1}{3}\left(-\frac{2}{3}\right) \\ &= 0\end{aligned}$$

$$\begin{aligned}
\langle \mathbf{u}_2, \mathbf{u}_3 \rangle &= \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3} \right) \cdot \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right) \\
&= \left(\frac{2}{3} \right) \left(\frac{1}{3} \right) + \left(\frac{1}{3} \right) \left(\frac{2}{3} \right) + \left(-\frac{2}{3} \right) \left(\frac{2}{3} \right) \\
&= 0 \\
\langle \mathbf{u}_3, \mathbf{u}_1 \rangle &= \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right) \cdot \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \right) \\
&= \left(\frac{1}{3} \right) \left(\frac{2}{3} \right) + \left(\frac{2}{3} \right) \left(-\frac{2}{3} \right) + \left(\frac{2}{3} \right) \left(\frac{1}{3} \right) \\
&= 0
\end{aligned}$$

$$\|\mathbf{u}_1\| = \langle \mathbf{u}_1, \mathbf{u}_1 \rangle^{\frac{1}{2}} = \left[\left(\frac{2}{3} \right)^2 + \left(-\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right]^{\frac{1}{2}} = 1$$

$$\|\mathbf{u}_2\| = \langle \mathbf{u}_2, \mathbf{u}_2 \rangle^{\frac{1}{2}} = \left[\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 + \left(-\frac{2}{3} \right)^2 \right]^{\frac{1}{2}} = 1$$

$$\|\mathbf{u}_3\| = \langle \mathbf{u}_3, \mathbf{u}_3 \rangle^{\frac{1}{2}} = \left[\left(\frac{1}{3} \right)^2 + \left(\frac{2}{3} \right)^2 + \left(\frac{2}{3} \right)^2 \right]^{\frac{1}{2}} = 1$$

Since $\langle \mathbf{u}_1, \mathbf{u}_2 \rangle = \langle \mathbf{u}_2, \mathbf{u}_3 \rangle = \langle \mathbf{u}_3, \mathbf{u}_1 \rangle = 0$ and $\|\mathbf{u}_1\| = \|\mathbf{u}_2\| = \|\mathbf{u}_3\| = 1$, the vectors are orthonormal.

Example 2: Show that the set of vectors $\mathbf{u}_1 = \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right)$, $\mathbf{u}_2 = \left(-\frac{1}{2}, \frac{1}{2}, 0\right)$, $\mathbf{u}_3 = \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right)$ is orthogonal with respect to the Euclidean inner product on R^3 and then convert it to an orthonormal set by normalizing the vectors.

Solution:

$$\begin{aligned}\langle \mathbf{u}_1, \mathbf{u}_2 \rangle &= \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right) \cdot \left(-\frac{1}{2}, \frac{1}{2}, 0\right) \\ &= \left(\frac{1}{5}\right)\left(-\frac{1}{2}\right) + \left(\frac{1}{5}\right)\left(\frac{1}{2}\right) + \left(\frac{1}{5}\right)(0) \\ &= 0\end{aligned}$$
$$\begin{aligned}\langle \mathbf{u}_2, \mathbf{u}_3 \rangle &= \left(-\frac{1}{2}, \frac{1}{2}, 0\right) \cdot \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right) \\ &= \left(-\frac{1}{2}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) + (0)\left(-\frac{2}{3}\right) \\ &= 0\end{aligned}$$
$$\begin{aligned}\langle \mathbf{u}_3, \mathbf{u}_1 \rangle &= \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}\right) \cdot \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}\right) \\ &= \left(\frac{1}{3}\right)\left(\frac{1}{5}\right) + \left(\frac{1}{3}\right)\left(\frac{1}{5}\right) + \left(-\frac{2}{3}\right)\left(\frac{1}{5}\right) \\ &= 0\end{aligned}$$

$$\|\mathbf{u}_1\| = \sqrt{\left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2} = \sqrt{\frac{3}{25}} = \frac{\sqrt{3}}{5}$$

$$\|\mathbf{u}_2\| = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{1}{2}}$$

$$\|\mathbf{u}_3\| = \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(-\frac{2}{3}\right)^2} = \sqrt{\frac{2}{3}}$$

Normalising the vectors,

$$\mathbf{w}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$$

$$\mathbf{w}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$$

$$\mathbf{w}_3 = \frac{\mathbf{u}_3}{\|\mathbf{u}_3\|} = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\sqrt{\frac{2}{3}} \right)$$

Orthonormal set = $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$

EXAMPLE 7.7 Find a nonzero vector w that is orthogonal to $u_1 = (1, 2, 1)$ and $u_2 = (2, 5, 4)$ in $\mathbb{R}^3$.

Let $w = (x, y, z)$. Then we want $\langle u_1, w \rangle = 0$ and $\langle u_2, w \rangle = 0$. This yields the homogeneous system

$$\begin{array}{rcl} x + 2y + z = 0 & \text{or} & x + 2y + z = 0 \\ 2x + 5y + 4z = 0 & & y + 2z = 0 \end{array}$$

Here z is the only free variable in the echelon system. Set $z = 1$ to obtain $y = -2$ and $x = 3$. Thus, $w = (3, -2, 1)$ is a desired nonzero vector orthogonal to u_1 and u_2 .

Any multiple of w will also be orthogonal to u_1 and u_2 . Normalizing w , we obtain the following unit vector orthogonal to u_1 and u_2 :

$$\hat{w} = \frac{w}{\|w\|} = \left(\frac{3}{\sqrt{14}}, -\frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}} \right)$$

EXAMPLE 7.9

- (a) Let $E = \{e_1, e_2, e_3\} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ be the usual basis of Euclidean space $\mathbf{R}^3$. It is clear that

$$\langle e_1, e_2 \rangle = \langle e_1, e_3 \rangle = \langle e_2, e_3 \rangle = 0 \quad \text{and} \quad \langle e_1, e_1 \rangle = \langle e_2, e_2 \rangle = \langle e_3, e_3 \rangle = 1$$

Namely, E is an orthonormal basis of $\mathbf{R}^3$. More generally, the usual basis of $\mathbf{R}^n$ is orthonormal for every n .

- (b) Let $V = C[-\pi, \pi]$ be the vector space of continuous functions on the interval $-\pi \leq t \leq \pi$ with inner product defined by $\langle f, g \rangle = \int_{-\pi}^{\pi} f(t)g(t) dt$. Then the following is a classical example of an orthogonal set in V :

$$\{1, \cos t, \cos 2t, \cos 3t, \dots, \sin t, \sin 2t, \sin 3t, \dots\}$$

4.3 ORTHOGONAL AND ORTHONORMAL BASIS

In an inner product space, a basis consisting of orthogonal vectors is called an orthogonal basis.

In an inner product space, a basis consisting of orthonormal vectors is called an orthonormal basis.

Theorem 4.1: If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an orthogonal set of non-zero vectors in an inner product space V then S is linearly independent.

Theorem 4.2: Any orthogonal set of n non-zero vectors in R^n is a basis for R^n .

Theorem 4.3: If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an orthonormal basis for an inner product space V , and $\mathbf{u}$ is any vector in V then it can be expressed as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$.

$$\mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \dots + \langle \mathbf{u}, \mathbf{v}_n \rangle \mathbf{v}_n$$

Here, $\langle \mathbf{u}, \mathbf{v}_1 \rangle, \langle \mathbf{u}, \mathbf{v}_2 \rangle, \dots, \langle \mathbf{u}, \mathbf{v}_n \rangle$ are the coordinate vectors of $\mathbf{u}$ with respect to the orthonormal basis S ,

i.e.,
$$[\mathbf{u}]_S = (\langle \mathbf{u}, \mathbf{v}_1 \rangle, \langle \mathbf{u}, \mathbf{v}_2 \rangle, \dots, \langle \mathbf{u}, \mathbf{v}_n \rangle)$$

Corollary: If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an orthogonal basis for an inner product space V , and $\mathbf{u}$ is any vector in V then

$$\mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{u}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 + \dots + \frac{\langle \mathbf{u}, \mathbf{v}_n \rangle}{\|\mathbf{v}_n\|^2} \mathbf{v}_n$$

Theorem 4.4: If S is an orthonormal basis for an n -dimensional inner product space, and if coordinate vectors of $\mathbf{u}$ and $\mathbf{v}$ with respect to S are $[\mathbf{u}]_S = (a_1, a_2, \dots, a_n)$ and $[\mathbf{v}]_S = (b_1, b_2, \dots, b_n)$ then,

- (i) $\|\mathbf{u}\| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$
- (ii) $d(\mathbf{u}, \mathbf{v}) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \dots + (a_n - b_n)^2}$
- (iii) $\langle \mathbf{u}, \mathbf{v} \rangle = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = [\mathbf{u}]_S \cdot [\mathbf{v}]_S$

Orthogonal Basis and Linear Combinations, Fourier Coefficients

Let S consist of the following three vectors in $\mathbf{R}^3$:

$$u_1 = (1, 2, 1), \quad u_2 = (2, 1, -4), \quad u_3 = (3, -2, 1)$$

The reader can verify that the vectors are orthogonal; hence, they are linearly independent. Thus, S is an orthogonal basis of $\mathbf{R}^3$.

Suppose we want to write $v = (7, 1, 9)$ as a linear combination of u_1, u_2, u_3 . First we set v as a linear combination of u_1, u_2, u_3 using unknowns x_1, x_2, x_3 as follows:

$$v = x_1 u_1 + x_2 u_2 + x_3 u_3 \quad \text{or} \quad (7, 1, 9) = x_1(1, 2, 1) + x_2(2, 1, -4) + x_3(3, -2, 1) \quad (*)$$

We can proceed in two ways.

METHOD 1: Expand $(*)$ (as in Chapter 3) to obtain the system

$$x_1 + 2x_2 + 3x_3 = 7, \quad 2x_1 + x_2 - 2x_3 = 1, \quad x_1 - 4x_2 + x_3 = 7$$

Solve the system by Gaussian elimination to obtain $x_1 = 3$, $x_2 = -1$, $x_3 = 2$. Thus, $v = 3u_1 - u_2 + 2u_3$.

Projections

Let V be an inner product space. Suppose w is a given nonzero vector in V , and suppose v is another vector. We seek the “projection of v along w ,” which, as indicated in Fig. 7-3(a), will be the multiple cw of w such that $v' = v - cw$ is orthogonal to w . This means

$$\langle v - cw, w \rangle = 0 \quad \text{or} \quad \langle v, w \rangle - c\langle w, w \rangle = 0 \quad \text{or} \quad c = \frac{\langle v, w \rangle}{\langle w, w \rangle}$$

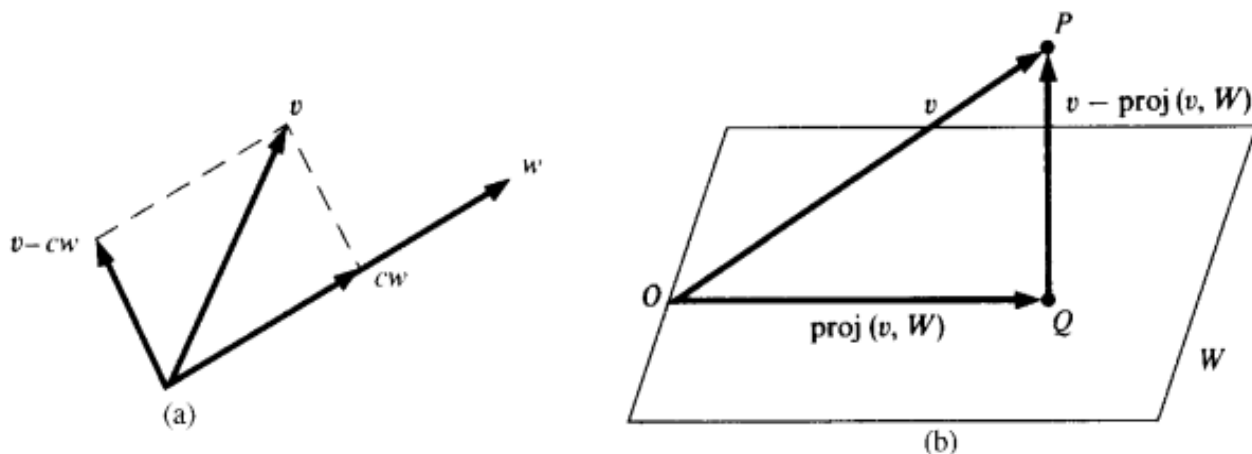


Figure 7-3

Accordingly, the *projection of v along w* is denoted and defined by

$$\text{proj}(v, w) = cw = \frac{\langle v, w \rangle}{\langle w, w \rangle} w$$

Such a scalar c is unique, and it is called the *Fourier coefficient* of v with respect to w or the *component* of v along w .

Example 1: Find the orthogonal projection of $\mathbf{u} = (1, -2, 3)$ along $\mathbf{v} = (1, 2, 1)$ in R^3 with respect to the Euclidean inner product.

Solution: Let W be the subspace spanned by the vector $\mathbf{v} = (1, 2, 1)$.

$$\begin{aligned}\text{proj}_W \mathbf{u} &= \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \mathbf{v} \\ &= \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v} \\ &= \frac{(1, -2, 3) \cdot (1, 2, 1)}{(1^2 + 2^2 + 1^2)} (1, 2, 1) \\ &= \frac{1 - 4 + 3}{6} (1, 2, 1) \\ &= 0\end{aligned}$$

7.7 Gram–Schmidt Orthogonalization Process

Suppose $\{v_1, v_2, \dots, v_n\}$ is a basis of an inner product space V . One can use this basis to construct an orthogonal basis $\{w_1, w_2, \dots, w_n\}$ of V as follows. Set

$$w_1 = v_1$$

$$w_2 = v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1$$

$$w_3 = v_3 - \frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2$$

.....

$$w_n = v_n - \frac{\langle v_n, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_n, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2 - \dots - \frac{\langle v_n, w_{n-1} \rangle}{\langle w_{n-1}, w_{n-1} \rangle} w_{n-1}$$

In other words, for $k = 2, 3, \dots, n$, we define

$$w_k = v_k - c_{k1}w_1 - c_{k2}w_2 - \dots - c_{k,k-1}w_{k-1}$$

where $c_{ki} = \langle v_k, w_i \rangle / \langle w_i, w_i \rangle$ is the component of v_k along w_i . By Theorem 7.8, each w_k is orthogonal to the preceding w 's. Thus, $w_1, w_2, \dots, w_n$ form an orthogonal basis for V as claimed. Normalizing each w_i will then yield an orthonormal basis for V .

The above construction is known as the *Gram–Schmidt orthogonalization process*. The following remarks are in order.

EXAMPLE 7.10 Apply the Gram–Schmidt orthogonalization process to find an orthogonal basis and then an orthonormal basis for the subspace U of $\mathbf{R}^4$ spanned by

$$v_1 = (1, 1, 1, 1), \quad v_2 = (1, 2, 4, 5), \quad v_3 = (1, -3, -4, -2)$$

(1) First set $w_1 = v_1 = (1, 1, 1, 1)$.

(2) Compute

$$v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 = v_2 - \frac{12}{4} w_1 = (-2, -1, 1, 2)$$

Set $w_2 = (-2, -1, 1, 2)$.

(3) Compute

$$v_3 - \frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2 = v_3 - \frac{(-8)}{4} w_1 - \frac{(-7)}{10} w_2 = \left(\frac{8}{5}, -\frac{17}{10}, -\frac{13}{10}, \frac{7}{5}\right)$$

Clear fractions to obtain $w_3 = (-6, -17, -13, 14)$.

Thus, w_1, w_2, w_3 form an orthogonal basis for U . Normalize these vectors to obtain an orthonormal basis $\{u_1, u_2, u_3\}$ of U . We have $\|w_1\|^2 = 4$, $\|w_2\|^2 = 10$, $\|w_3\|^2 = 910$, so

$$u_1 = \frac{1}{2}(1, 1, 1, 1), \quad u_2 = \frac{1}{\sqrt{10}}(-2, -1, 1, 2), \quad u_3 = \frac{1}{\sqrt{910}}(16, -17, -13, 14)$$

EXAMPLE 7.11 Let V be the vector space of polynomials $f(t)$ with inner product $\langle f, g \rangle = \int_{-1}^1 f(t)g(t) dt$. Apply the Gram–Schmidt orthogonalization process to $\{1, t, t^2, t^3\}$ to find an orthogonal basis $\{f_0, f_1, f_2, f_3\}$ with integer coefficients for $\mathbf{P}_3(t)$.

Here we use the fact that, for $r + s = n$,

$$\langle t^r, t^s \rangle = \int_{-1}^1 t^n dt = \frac{t^{n+1}}{n+1} \Big|_{-1}^1 = \begin{cases} 2/(n+1) & \text{when } n \text{ is even} \\ 0 & \text{when } n \text{ is odd} \end{cases}$$

- (1) First set $f_0 = 1$.
- (2) Compute $t = \frac{\langle t, 1 \rangle}{\langle 1, 1 \rangle} (1) = t - 0 = t$. Set $f_1 = t$.
- (3) Compute

$$t^2 - \frac{\langle t^2, 1 \rangle}{\langle 1, 1 \rangle} (1) - \frac{\langle t^2, t \rangle}{\langle t, t \rangle} (t) = t^2 - \frac{2}{2}(1) + 0(t) = t^2 - \frac{1}{3}$$

Multiply by 3 to obtain $f_2 = 3t^2 - 1$.

(4) Compute

$$\begin{aligned} t^3 - \frac{\langle t^3, 1 \rangle}{\langle 1, 1 \rangle} (1) - \frac{\langle t^3, t \rangle}{\langle t, t \rangle} (t) - \frac{\langle t^3, 3t^2 - 1 \rangle}{\langle 3t^2 - 1, 3t^2 - 1 \rangle} (3t^2 - 1) \\ = t^3 - 0(1) - \frac{2}{\frac{5}{3}}(t) - 0(3t^2 - 1) = t^3 - \frac{3}{5}t \end{aligned}$$

Multiply by 5 to obtain $f_3 = 5t^3 - 3t$.

Thus, $\{1, t, 3t^2 - 1, 5t^3 - 3t\}$ is the required orthogonal basis.

Remark: Normalizing the polynomials in Example 7.11 so that $p(1) = 1$ yields the polynomials

$$1, t, \frac{1}{2}(3t^2 - 1), \frac{1}{2}(5t^3 - 3t)$$

These are the first four *Legendre polynomials*, which appear in the study of differential equations.

► **EXAMPLE 8 Using the Gram–Schmidt Process**

Assume that the vector space R^3 has the Euclidean inner product. Apply the Gram–Schmidt process to transform the basis vectors

$$\mathbf{u}_1 = (1, 1, 1), \quad \mathbf{u}_2 = (0, 1, 1), \quad \mathbf{u}_3 = (0, 0, 1)$$

into an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, and then normalize the orthogonal basis vectors to obtain an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$.

Solution

Step 1. $\mathbf{v}_1 = \mathbf{u}_1 = (1, 1, 1)$

Step 2. $\mathbf{v}_2 = \mathbf{u}_2 - \text{proj}_{W_1} \mathbf{u}_2 = \mathbf{u}_2 - \frac{\langle \mathbf{u}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1$

$$= (0, 1, 1) - \frac{2}{3}(1, 1, 1) = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

Step 3. $\mathbf{v}_3 = \mathbf{u}_3 - \text{proj}_{W_2} \mathbf{u}_3 = \mathbf{u}_3 - \frac{\langle \mathbf{u}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{u}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2$

$$= (0, 0, 1) - \frac{1}{3}(1, 1, 1) - \frac{1/3}{2/3} \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$
$$= \left(0, -\frac{1}{2}, \frac{1}{2}\right)$$

$$\mathbf{v}_1 = (1, 1, 1), \quad \mathbf{v}_2 = \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right), \quad \mathbf{v}_3 = \left(0, -\frac{1}{2}, \frac{1}{2}\right)$$

form an orthogonal basis for R^3 . The norms of these vectors are

$$\|\mathbf{v}_1\| = \sqrt{3}, \quad \|\mathbf{v}_2\| = \frac{\sqrt{6}}{3}, \quad \|\mathbf{v}_3\| = \frac{1}{\sqrt{2}}$$

so an orthonormal basis for R^3 is

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \quad \mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right),$$

$$\mathbf{q}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \quad \blacktriangleleft$$

Matrix Representation of an Inner Product

Let $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a basis of an n -dimensional inner product space V . For vectors $\mathbf{u}, \mathbf{v} \in V$, write

$$\mathbf{u} = x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + \cdots + x_n\mathbf{u}_n,$$

$$\mathbf{v} = y_1\mathbf{u}_1 + y_2\mathbf{u}_2 + \cdots + y_n\mathbf{u}_n.$$

The linearity implies

$$\begin{aligned}\langle \mathbf{u}, \mathbf{v} \rangle &= \left\langle \sum_{i=1}^n x_i \mathbf{u}_i, \sum_{j=1}^n y_j \mathbf{u}_j \right\rangle \\ &= \sum_{i=1}^n \sum_{j=1}^n x_i y_j \langle \mathbf{u}_i, \mathbf{u}_j \rangle.\end{aligned}$$

We call the $n \times n$ matrix

$$A = \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{u}_1 \rangle & \langle \mathbf{u}_1, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_1, \mathbf{u}_n \rangle \\ \langle \mathbf{u}_2, \mathbf{u}_1 \rangle & \langle \mathbf{u}_2, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_2, \mathbf{u}_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{u}_n, \mathbf{u}_1 \rangle & \langle \mathbf{u}_n, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_n \rangle \end{bmatrix}$$

the **matrix of the inner product $\langle, \rangle$ relative to the basis $\mathcal{B}$** . Thus, using coordinate vectors

$$[\mathbf{u}]_{\mathcal{B}} = [x_1, x_2, \dots, x_n]^T, \quad [\mathbf{v}]_{\mathcal{B}} = [y_1, y_2, \dots, y_n]^T,$$

we have

$$\langle \mathbf{u}, \mathbf{v} \rangle = [\mathbf{u}]_{\mathcal{B}}^T A [\mathbf{v}]_{\mathcal{B}}.$$

4 Representation of inner product

Theorem 4.1. *Let V be an n -dimensional vector space with an inner product $\langle, \rangle$, and let A be the matrix of $\langle, \rangle$ relative to a basis $\mathcal{B}$. Then for any vectors $\mathbf{u}, \mathbf{v} \in V$,*

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{x}^T A \mathbf{y}.$$

where $\mathbf{x}$ and $\mathbf{y}$ are the coordinate vectors of $\mathbf{u}$ and $\mathbf{v}$, respectively, i.e., $\mathbf{x} = [\mathbf{u}]_{\mathcal{B}}$ and $\mathbf{y} = [\mathbf{v}]_{\mathcal{B}}$.

Example 4.1. For the inner product of $\mathbb{R}^3$ defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle = 2x_1y_1 - x_1y_2 - x_2y_1 + 5x_2y_2,$$

where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \in \mathbb{R}^2$, its matrix relative to the standard basis $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2\}$ is

$$A = \begin{bmatrix} \langle \mathbf{e}_1, \mathbf{e}_1 \rangle & \langle \mathbf{e}_1, \mathbf{e}_2 \rangle \\ \langle \mathbf{e}_2, \mathbf{e}_1 \rangle & \langle \mathbf{e}_2, \mathbf{e}_2 \rangle \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 5 \end{bmatrix}.$$

The inner product can be written as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T A \mathbf{y} = [x_1, x_2] \begin{bmatrix} 2 & -1 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.$$

Let V be a real inner product space with basis $S = \{u_1, u_2, \dots, u_n\}$. The matrix

$$A = [a_{ij}], \quad \text{where} \quad a_{ij} = \langle u_i, u_j \rangle$$

is called the *matrix representation of the inner product on V relative to the basis S* .

Observe that A is symmetric, because the inner product is symmetric; that is, $\langle u_i, u_j \rangle = \langle u_j, u_i \rangle$. Also, A depends on both the inner product on V and the basis S for V . Moreover, if S is an orthogonal basis, then A is diagonal, and if S is an orthonormal basis, then A is the identity matrix.

EXAMPLE 7.14 The vectors $u_1 = (1, 1, 0)$, $u_2 = (1, 2, 3)$, $u_3 = (1, 3, 5)$ form a basis S for Euclidean space $\mathbf{R}^3$. Find the matrix A that represents the inner product in $\mathbf{R}^3$ relative to this basis S .

First compute each $\langle u_i, u_j \rangle$ to obtain

$$\begin{aligned} \langle u_1, u_1 \rangle &= 1 + 1 + 0 = 2, & \langle u_1, u_2 \rangle &= 1 + 2 + 0 = 3, & \langle u_1, u_3 \rangle &= 1 + 3 + 0 = 4 \\ \langle u_2, u_2 \rangle &= 1 + 4 + 9 = 14, & \langle u_2, u_3 \rangle &= 1 + 6 + 15 = 22, & \langle u_3, u_3 \rangle &= 1 + 9 + 25 = 35 \end{aligned}$$

Then $A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 14 & 22 \\ 4 & 22 & 35 \end{bmatrix}$. As expected, A is symmetric.

Orthogonal Matrices

- 7.79.** Find the number and exhibit all 2×2 orthogonal matrices of the form $\begin{bmatrix} \frac{1}{3} & x \\ y & z \end{bmatrix}$.
- 7.80.** Find a 3×3 orthogonal matrix P whose first two rows are multiples of $u = (1, 1, 1)$ and $v = (1, -2, 3)$, respectively.
- 7.81.** Find a symmetric orthogonal matrix P whose first row is $(\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$. (Compare with Problem 7.32.)
- 7.82.** Real matrices A and B are said to be *orthogonally equivalent* if there exists an orthogonal matrix P such that $B = P^T A P$. Show that this relation is an equivalence relation.

Positive Definite Matrices and Inner Products

- 7.83.** Find the matrix A that represents the usual inner product on $\mathbf{R}^2$ relative to each of the following bases:

(a) $\{v_1 = (1, 4), v_2 = (2, -3)\}$, (b) $\{w_1 = (1, -3), w_2 = (6, 2)\}$.

- 7.84.** Consider the following inner product on $\mathbf{R}^2$:

$$f(u, v) = x_1 y_1 - 2x_1 y_2 - 2x_2 y_1 + 5x_2 y_2, \quad \text{where} \quad u = (x_1, x_2) \quad v = (y_1, y_2)$$

Find the matrix B that represents this inner product on $\mathbf{R}^2$ relative to each basis in Problem 7.83.

- 7.85.** Find the matrix C that represents the usual basis on $\mathbf{R}^3$ relative to the basis S of $\mathbf{R}^3$ consisting of the vectors $u_1 = (1, 1, 1)$, $u_2 = (1, 2, 1)$, $u_3 = (1, -1, 3)$.

- 7.86.** Let $V = \mathbf{P}_2(t)$ with inner product $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$.

- (a) Find $\langle f, g \rangle$, where $f(t) = t + 2$ and $g(t) = t^2 - 3t + 4$.
- (b) Find the matrix A of the inner product with respect to the basis $\{1, t, t^2\}$ of V .
- (c) Verify Theorem 7.16 that $\langle f, g \rangle = [f]^T A [g]$ with respect to the basis $\{1, t, t^2\}$.